

A SHARP LOWER BOUND FOR THREE-NEIGHBOR BOOTSTRAP PERCOLATION ON FINITE RECTANGLES

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ABSTRACT. We determine a sharp lower bound for deterministic bootstrap percolation with threshold three on an $m \times n$ rectangle in the square lattice. If an initial set percolates, then its cardinality is at least

$$\frac{mn + m + n}{3}.$$

The proof is based on a simple invariant involving the number of infected sites, adjacent infected pairs, and bounded components of the complement. We also classify the extremal configurations. Equality holds precisely for a self-similar family of square configurations, which we call quincunxes; equivalently, equality is possible only when $m = n = 2^k - 1$.

1. INTRODUCTION

Bootstrap percolation is a deterministic cellular automaton introduced in statistical physics [3] and subsequently studied extensively in probability and combinatorics; see, for example, [1, 4, 2]. Given a graph G , a threshold r , and an initial infected set $S \subseteq V(G)$, vertices are added to S whenever they have at least r infected neighbors. The usual probabilistic problem asks when a random initial set percolates. In this paper we consider an extremal finite problem: how small can a percolating set be?

Let $R_{m,n} = [m] \times [n]$ be the $m \times n$ rectangular grid, with nearest-neighbor adjacency. We study the threshold-three rule: a site becomes infected once at least three of its four lattice neighbors are infected. Boundary sites have fewer than four possible neighbors, so this rule is especially restrictive near the boundary.

Our main result is the following.

Theorem 1.1 (Main theorem). *Let A be a percolating 0-1 matrix of shape $m \times n$ for the threshold-three bootstrap process. Then*

$$(1) \quad \text{wt}(A) \geq \frac{mn + m + n}{3}.$$

Moreover, equality holds if and only if $m = n = 2^k - 1$ for some $k \geq 1$ and $A = Q_k$, where Q_k is the quincunx matrix defined in Section 4.

The problem considered here may be viewed as a threshold-three analogue of a finite infected-checkerboard puzzle popularized by Winkler [5]. The lower bound is obtained from a conserved quantity. The equality case is considerably more rigid: an extremal configuration has no adjacent infected pair and no bounded component in its complement. These two prohibitions force an alternating boundary, a parity condition, and finally a recursive self-similar structure.

2. THE PROCESS AND ITS CLOSURE

We write

$$R_{m,n} = [m] \times [n], \quad \text{where } [m] = \{1, 2, \dots, m\}.$$

A *site* is an element of $R_{m,n}$. Two sites are *adjacent* if their ℓ^1 -distance is one, and *weakly adjacent* if their ℓ^∞ -distance is one. Thus weak adjacency allows diagonal steps. A site is a *boundary site* if it lies in the first or last row or in the first or last column.

Let A be a 0-1 matrix on $R_{m,n}$. We write

$$\text{supp}(A) = \{x \in R_{m,n} : A_x = 1\}, \quad Z(A) = \{x \in R_{m,n} : A_x = 0\}.$$

The *weight* of A is $\text{wt}(A) = |\text{supp}(A)|$. A site with value 0 is *eligible* if at least three of its adjacent sites have value 1. The threshold-three bootstrap process repeatedly changes eligible sites from value 0 to value 1; sites with value 1 never change. We say that A *percolates* if the process eventually reaches the all-one matrix.

The final state is independent of whether eligible sites are added simultaneously or one at a time. For a set $S \subseteq R_{m,n}$, define

$$T(S) = S \cup \{x \in R_{m,n} : x \text{ has at least three neighbors in } S\}.$$

Since T is monotone, the increasing sequence

$$S \subseteq T(S) \subseteq T^2(S) \subseteq \dots$$

stabilizes. Its stable value is the *closure* of S , denoted $\bar{S}$. Equivalently, $\bar{S}$ is the unique minimal superset of S that is closed under the threshold-three rule. Thus A percolates if and only if $\overline{\text{supp}(A)} = R_{m,n}$.

3. THE INVARIANT AND THE LOWER BOUND

A *link* of A is an unordered adjacent pair of sites in $\text{supp}(A)$. Let $\text{links}(A)$ denote the number of links. A *void* of A is a connected component of $Z(A)$, using ordinary adjacency, that does not meet the boundary of $R_{m,n}$. Let $\text{voids}(A)$ denote the number of voids.

Proposition 3.1 (The invariant). *The quantity*

$$(2) \quad \chi(A) = 3 \text{wt}(A) - \text{links}(A) - \text{voids}(A)$$

is invariant under the threshold-three bootstrap process.

Proof. It is enough to consider the addition of a single eligible site x . The weight increases by one. If exactly three neighbors of x are infected, then three new links are created. In this case the number of voids is unchanged. Indeed, if x is an interior site, then it has exactly one zero-neighbor before it changes, so deleting x from its zero-component removes a leaf and neither creates nor destroys a bounded zero-component. If x is a boundary site, then all of its neighbors are infected, and its zero-component is the singleton $\{x\}$, which is not a void.

If four neighbors of x are infected, then x is an isolated interior zero. Four new links are created, and exactly one void, the singleton $\{x\}$, disappears. In both cases the net change in $3 \text{wt} - \text{links} - \text{voids}$ is zero. $\square$

Corollary 3.2 (The lower bound). *If A percolates on $R_{m,n}$, then*

$$\text{wt}(A) \geq \frac{mn + m + n}{3}.$$

Equality holds if and only if A has no links and no voids.

Proof. Let J be the all-one matrix on $R_{m,n}$. Since A percolates, Proposition 3.1 gives $\chi(A) = \chi(J)$. Now

$$\text{wt}(J) = mn, \quad \text{links}(J) = m(n-1) + (m-1)n = 2mn - m - n, \quad \text{voids}(J) = 0.$$

Therefore

$$(3) \quad 3 \operatorname{wt}(A) - \operatorname{links}(A) - \operatorname{voids}(A) = mn + m + n,$$

and hence

$$\operatorname{wt}(A) = \frac{mn + m + n + \operatorname{links}(A) + \operatorname{voids}(A)}{3}.$$

The bound and the equality criterion follow immediately. $\square$

We shall call a percolating matrix *perfect* if it has no links and no voids. By Corollary 3.2, perfect matrices are exactly the equality cases in (1).

4. QUINCUNXES

Let $\nu_2(t)$ denote the exponent of the largest power of 2 dividing the positive integer t . For $k \geq 1$, define Q_k to be the square 0-1 matrix of order $2^k - 1$ given by

$$(4) \quad Q_k(i, j) = 1 \iff \nu_2(i) = \nu_2(j).$$

We call Q_k the *k-th quincunx matrix*.

The closed form immediately implies the recursion

$$(5) \quad Q_1 = (1), \quad Q_{k+1} = \begin{pmatrix} Q_k & 0 & Q_k \\ 0 & 1 & 0 \\ Q_k & 0 & Q_k \end{pmatrix}.$$

Equivalently, for $k \geq 1$, the matrix Q_{k+1} is the square matrix of order $2^{k+1} - 1$ whose entries are given by $Q_{k+1}(2i, 2j) = Q_k(i, j)$, $Q_{k+1}(i, j) = 1$ if i and j are both odd, and $Q_{k+1}(i, j) = 0$ if i and j have opposite parity. Thus Q_k is recovered from Q_{k+1} by deleting all rows and columns of odd index.

Proposition 4.1. *Every quincunx matrix is perfect.*

Proof. First we show that Q_k percolates. This is trivial for $k = 1$. Suppose that Q_k percolates to the all-one matrix J_k . By the recursion (5), the four corner blocks of Q_{k+1} are copies of Q_k , and hence each percolates to J_k . At that point the central row and central column fill outward from the center: each next zero in one of these arms has the center-side neighbor infected and has two infected neighbors in the adjacent filled blocks. Thus Q_{k+1} percolates.

It remains to verify equality in the lower bound. From (4), or directly from (5),

$$\operatorname{wt}(Q_1) = 1, \quad \operatorname{wt}(Q_{k+1}) = 4 \operatorname{wt}(Q_k) + 1.$$

Hence

$$(6) \quad \operatorname{wt}(Q_k) = \frac{4^k - 1}{3}.$$

Since Q_k has order $2^k - 1$, this is precisely

$$\frac{(2^k - 1)^2 + 2(2^k - 1)}{3}.$$

Thus Q_k attains equality in Corollary 3.2, and is perfect. $\square$

5. EXTREMAL RIGIDITY

We now prove the converse: every perfect matrix is a quincunx. We begin with three local obstructions.

Lemma 5.1 (Local obstructions). *A percolating matrix cannot have any of the following configurations in its zero set:*

- (i) a corner site;
- (ii) two adjacent boundary sites;
- (iii) a 2×2 block of sites.

Proof. In each case, suppose for contradiction that all sites in the displayed set are initially zero, and consider the first of them to become infected. At that moment all the others in the set are still zero. In cases (i) and (ii), the chosen site has at most two possible infected neighbors; in case (iii), it has two zero-neighbors inside the 2×2 block and hence again at most two infected neighbors. This contradicts the threshold-three rule. $\square$

Lemma 5.2 (Boundary alternation). *Let A be a perfect matrix with at least two rows and at least two columns. Then the boundary values alternate between 0 and 1, starting with value 1 at every corner. In particular, both dimensions of A are odd.*

Proof. The corner sites have value 1 by Lemma 5.1. Since A has no links, no two adjacent boundary sites can both have value 1. Since A percolates, Lemma 5.1 forbids two adjacent boundary sites from both having value 0. Thus the boundary values alternate. Since all four corners have value 1, both side lengths are odd. $\square$

Lemma 5.3 (Weak connectivity). *If A percolates, then $\text{supp}(A)$ is weakly connected.*

Proof. It is enough to prove that adding a single eligible site does not change the number of weak components of the support. Let x be an eligible site. Its infected neighbors form a weakly connected set: among any three of the four orthogonal neighbors of x , consecutive neighbors around x are weakly adjacent.

Thus all infected neighbors of x lie in a single weak component C . When x is added, it is weakly adjacent to this component, so it does not create a new component. It also cannot merge two old components. Indeed, let $y \in \text{supp}(A)$ be weakly adjacent to x . If y is an orthogonal neighbor of x , then $y \in C$. If y is diagonal from x , then it is weakly adjacent to two orthogonal neighbors of x , at least one of which is infected, since at most one orthogonal neighbor of x is not infected. Hence y was already weakly connected to C . Therefore no distinct weak components are merged.

The number of weak components is therefore invariant during the process. The final support is all of $R_{m,n}$, which is weakly connected, so the initial support was weakly connected. $\square$

Lemma 5.4 (Parity). *If A is perfect, then every site of $\text{supp}(A)$ has even parity.*

Proof. By Lemma 5.3, the support is weakly connected. Since A has no links, two weakly adjacent infected sites cannot be horizontally or vertically adjacent; each weak step within the support is therefore diagonal. Diagonal steps preserve the parity of the sum of the coordinates. Hence all sites in $\text{supp}(A)$ have the same parity. A corner site has value 1 by Lemma 5.2, and the corner $(1, 1)$ has even parity. $\square$

Lemma 5.5 (Odd-odd sites). *If A is perfect, then every site (i, j) with i and j both odd has value 1.*

Proof. Let

$$S = \{(i, j) \in Z(A) : i \text{ and } j \text{ are both odd}\}.$$

Suppose, for contradiction, that $S \neq \emptyset$. Since A has no voids, some site of S is connected in $Z(A)$ to the boundary. Choose a path

$$z_0, z_1, \dots, z_r$$

in $Z(A)$, with $z_0 \in S$ and z_r on the boundary, such that r is minimal. By Lemma 5.2, no boundary site with two odd coordinates lies in $Z(A)$; hence z_0 is not a boundary site. Since z_0 has two odd coordinates, none of its orthogonal neighbors is a boundary site, so $r \geq 2$.

The four orthogonal neighbors of z_0 have odd parity, so by Lemma 5.4 they all have value 0. If a diagonal neighbor of z_0 also had value 0, then together with z_0 and the two intervening orthogonal neighbors it would form a zero 2×2 block, contrary to Lemma 5.1. Therefore all diagonal neighbors of z_0 have value 1.

The path may be chosen simple, so $z_2 \neq z_0$. Also z_2 cannot be diagonally adjacent to z_0 , since all diagonal neighbors of z_0 have value 1. Thus after two adjacent steps from z_0 , the site z_2 must be two horizontal steps or two vertical steps from z_0 . In particular, z_2 also has two odd coordinates. Since $z_2 \in Z(A)$, we have $z_2 \in S$. But then

$$z_2, z_3, \dots, z_r$$

is a shorter path in $Z(A)$ from a site of S to the boundary, contradicting the choice of the original path. $\square$

Proposition 5.6 (Self-similar reduction). *Let A be a perfect matrix of shape $(2p+1) \times (2q+1)$, and let B be the $p \times q$ submatrix defined by*

$$B_{x,y} = A_{2x,2y} \quad (1 \leq x \leq p, 1 \leq y \leq q).$$

Then B is perfect.

Proof. First we prove that B percolates. Choose a legal one-at-a-time percolation order for A , and restrict attention to the sites of A whose two coordinates are even. This induces an order on the sites of B .

Let $(x, y) \in Z(B)$, and put $c = (2x, 2y)$. When c becomes infected in the chosen percolation order for A , at least three of its four orthogonal neighbors have already become infected. Consider one such neighbor p_0 . By Lemma 5.4, the site p_0 was initially zero, so it became infected earlier than c . At the moment when p_0 became infected, the site c was still zero. Hence p_0 could not be a boundary site, and the site q_0 opposite c across p_0 must already have been infected. This opposite site has two even coordinates and corresponds to a neighbor of (x, y) in B . Applying this argument to three infected neighbors of c , we find that (x, y) has at least three neighbors in B that are earlier in the induced order or are initially infected in B . Thus the induced order is a legal percolation order for B . Hence B percolates.

It remains to compute its weight. By Lemmas 5.4 and 5.5, the support of A consists exactly of all sites with two odd coordinates, together with the even-even sites corresponding to $\text{supp}(B)$. Therefore

$$(7) \quad \text{wt}(A) = (p+1)(q+1) + \text{wt}(B).$$

Since A is perfect,

$$\text{wt}(A) = \frac{(2p+1)(2q+1) + (2p+1) + (2q+1)}{3}.$$

Substituting this into (7) gives

$$\text{wt}(B) = \frac{pq + p + q}{3}.$$

Since B percolates, equality in Corollary 3.2 shows that B is perfect. $\square$

Proposition 5.7 (Classification of perfect matrices). *A matrix is perfect if and only if it is a quincunx matrix Q_k for some $k \geq 1$.*

Proof. Proposition 4.1 shows that every quincunx is perfect. Conversely, let A be perfect. We prove by induction on the area of A that A is a quincunx.

If A has only one row or only one column, then no zero site can ever become infected. Thus a percolating such matrix must be all-one. Since a perfect matrix has no links, this forces $A = (1) = Q_1$.

Now suppose that A has at least two rows and at least two columns. By Lemma 5.2, its dimensions are odd, say $(2p+1) \times (2q+1)$. Let B be its even-even submatrix. By Proposition 5.6, B is perfect. The matrix B has smaller area, so by induction $B = Q_k$ for some k . In particular, $p = q = 2^k - 1$.

Lemmas 5.4 and 5.5 determine all entries of A : the sites with two odd coordinates have value 1, the sites with one odd and one even coordinate have value 0, and the even-even submatrix is $B = Q_k$. This is exactly the recursive construction of Q_{k+1} . Hence $A = Q_{k+1}$. $\square$

Proof of Theorem 1.1. The lower bound is Corollary 3.2. Equality holds exactly for perfect matrices, and these are classified by Proposition 5.7. Since Q_k has order $2^k - 1$, the stated equality condition follows. $\square$

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